Aarnav Singh

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Professional Summary

Graduate Trainee at HCLTech with a B.Tech in Computer Science & Engineering (Cybersecurity specialization) from SRM University. Built a working foundation in enterprise network security, perimeter defense, and cloud infrastructure across Azure, AWS, and GCP through a hands-on internship, structured professional training, and certifications including Google Cloud Cybersecurity and Fortinet NSE. Genuinely drawn to how networks are built and defended, and looking to grow across cybersecurity operations, network engineering, and enterprise IT security roles.

Education

SRM University India
B.Tech in Computer Science & Engineering with specialization in Cybersecurity *2022 – 2026*
- GPA: 7.74/10

Experience

HCLTech – Graduate Trainee August 2026 – Present
Noida, India
- Completing a structured pre-joining training program spanning cloud platforms (AWS, Azure, GCP, OCI), networking, Linux system administration, and DevOps/SRE practices (Terraform, Grafana, Prometheus, ITIL 4).
- Passed HCLTech's proctored pre-joining assessment, validating readiness across enterprise IT, cloud, and security fundamentals.

HCLTech – Intern May 2024 – July 2024
Noida, India
- Gained direct exposure to enterprise-scale Cybersecurity operations, developing a working understanding of how security is structured and coordinated within a large IT organization.
- Built foundational knowledge of server environments and enterprise networking, bridging classroom theory with applied, enterprise-scale context.
- Tested a Python-based automation script for provisioning Azure Virtual Machines, gaining hands-on exposure to cloud infrastructure automation.

Projects

Adversarial Attack Detection in Zero-Trust Networks Jan 2026
- *GitHub:* github.com/Aarnav-Singh/adversarial-ml-security-framework
- Built a six-layer Zero-Trust network security system in Python/PyTorch combining ML-based intrusion detection with a context-aware policy engine.
- Designed a five-decision policy engine (ALLOW / DENY / STEP_UP_AUTH / RATE_LIMIT / ISOLATE) that made context-aware, high-stakes decisions in real time, reducing abuse/evasion bypass rates from 65% to 2.2% without requiring model retraining.
- Identified gaps in existing detection policy and implemented targeted fixes, validating improvements through a 5-seed evaluation across two real-world datasets (UNSW-NB15, CICIDS-2017).
- Co-authored a research paper based on this work, currently under review at the Journal of Information Security and Applications (Elsevier).

Hybrid Behavioral Intrusion Detection System Feb 2026
- *GitHub:* github.com/Aarnav-Singh/Hybrid-Behavioral-Intrusion-Detection-System
- Designed and developed a full-stack hybrid intrusion detection system combining rule-based detection with Isolation Forest-based anomaly detection for behavioral threat analysis.
- Built a real-time React dashboard using WebSockets and REST APIs for live threat visualization and alerting.
- Integrated Elasticsearch for centralized log indexing and threat-event correlation, and deployed Prometheus for monitoring detection latency and system performance.
- Containerized the full system using Docker to simulate a scalable deployment environment aligned with Zero-Trust and modern security monitoring principles.

Technical Skills

Security & Networking: Networking Fundamentals (TCP/IP, DNS, Routing & Switching, VLANs/Subnetting) — Network Security Analysis — Threat Detection & Anomaly Analysis — Intrusion Detection Systems (IDS/IPS) Design — Zero-Trust Architecture & Policy Design — Adversarial ML & Evasion Analysis — Vulnerability Assessment — Log Analysis & Correlation — Wireshark — Nmap — Nessus — Metasploit — Kali Linux — FortiGate — Cisco IOS / Cisco Networking

Cloud & DevOps: Cloud Infrastructure Monitoring — IT Service Management (ITIL 4 Concepts) — Site Reliability Engineering (SRE) Fundamentals — Infrastructure as Code (IaC) — AWS — Microsoft Azure — Google Cloud Platform (GCP) — Terraform — Prometheus — Grafana — Docker — Linux (RHEL)

Programming, Data & Collaboration: Python — PyTorch — React.js — Elasticsearch — SQL/MySQL — Database Fundamentals (SQL/RDBMS) — Git/GitHub — Jira — Agile & Scrum Methodologies — GenAI/LLM Fundamentals & Prompt Engineering — Technical Documentation & Research Writing — Cross-Functional Collaboration — Windows

Certifications

Google Cloud Cybersecurity Certificate

Fortinet NSE 3 Certified in Cybersecurity

Fortinet NSE 2 Certified in Cybersecurity

Fortinet NSE 1 Certified in Cybersecurity

Cisco – Networking Basics

Cisco – Introduction to Cybersecurity

NPTEL – Advanced Computer Networks

Deloitte Australia (Forage) – Cyber Job Simulation

Leadership & Activities

Volunteer – Purvanchal Vikas Foundation June 2024 – July 2024
Noida, India
- Supported outreach initiatives and beneficiary enrollment programs.

Core Member & Event Host – Gamer's Creed Esports Club Sept 2024
Modinagar, India
- Hosted and coordinated college-level esports events.